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EMG-based wake gestures eliminate false activations during out-of-set activities of daily living: an online myoelectric control study

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E-mail: eeddy@unb.ca, evan.campbell1@unb.ca, scottb@unb.ca and escheme@unb.ca**Keywords:** electromyography, myoelectric control, false activations, online evaluation, wake gesture

Abstract

Objective. While myoelectric control has been commercialized in prosthetics for decades, its adoption for more general human–machine interaction has been slow. Although high accuracies can be achieved across many gestures, current control approaches are prone to false activations in real-world conditions. This is because the same electromyogram (EMG) signals generated during the elicitation of gestures are also naturally activated when performing activities of daily living (ADLs), such as when driving to work or while typing on a keyboard. This can lead the myoelectric control system, which is trained on a closed set of gestures and thus unaware of the muscle activity associated with these ADLs, to be falsely activated, leading to erroneous inputs and user frustration. **Approach.** To overcome this problem, the concept of wake gestures, whereby users could switch between a dedicated control mode and a sleep mode by snapping their fingers, was explored. Using a simple dynamic time warping model, the real-world user-in-the-loop efficacy of wake gestures as a toggle for myoelectric interfaces was demonstrated through two online ubiquitous control tasks with varying levels of difficulty: (1) dismissing an alarm and (2) controlling a robot. **Main results.** During these online evaluations, the designed system ignored almost all (>99.9%) non-target EMG activity generated during a set of ADLs (i.e. walking, typing, writing, phone use, and driving), ignored all control gestures (i.e. wrist flexion, wrist extension, hand open, and hand close), and enabled reliable mode switching during intentional wake gesture elicitation. Additionally, questionnaires revealed that participants responded well to the use of wake gestures and generally preferred false negatives over false positives, providing valuable insights into the future design of these systems. **Significance.** These results highlight the real-world viability of wake gestures for enabling the intermittent use of myoelectric control, opening up new interaction possibilities for EMG-based inputs.

1. Introduction

With new opportunities emerging with mixed reality, smart homes, and other connected technologies, modern-day human–machine interaction is rapidly evolving, leading to an increased need for novel interaction techniques. Myoelectric control, which involves the recognition of muscle-based inputs (or gestures) via the electromyogram (EMG) signals generated during muscular contractions, is an interesting

avenue due to its ability to work in scenarios where other technologies may not [1]. For example, compared to speech, it is more subtle and inconspicuous and could be used in public settings. Moreover, it does not have the line of sight requirement of computer vision systems, meaning gestures can be elicited anywhere, such as in a coat pocket and in environments with poor lighting conditions.

Historically, myoelectric control has evolved for the primary purpose of enabling amputees to control

powered prostheses [2–4]. A fundamental distinction between this traditional use case and more general human–machine interaction is the difference between *dedicated* and *intermittent* control. Dedicated control refers to a system that is always classifying EMG as input intent, often with an ‘other’ or ‘no movement’ class acting as a ‘do nothing’ command. Controlling a prosthetic device usually assumes dedicated control, where individuals constantly adjust their prosthesis based on predictions occurring on milliseconds worth of data [5]. In these cases, the classification problem is considered to be *closed-set*, where the inputs to the system are assumed to correspond to one of the N classes used for training. This assumption is seemingly appropriate in such settings because amputees primarily engage in dedicated single-task device control; that is, it is assumed that they only activate the muscles of their residual limb to control the prosthesis. Such an approach, however, does not necessarily work for a user of a more general myoelectric interface.

Many EMG researchers continue to develop dedicated myoelectric control systems, partially because solving the out-of-set problem is an ongoing machine-learning challenge [6]. For example, in the recent influential work by Ctrl Labs, all three cross-user models developed (1-D scrolling, discrete navigation, and handwriting recognition) were all for dedicated control tasks, so it is currently unknown if users may be limited in their ability to engage in other activities [7]. Moving forward, the use of myoelectric interfaces will likely be desirable in more intermittent contexts, such as to control a music player while walking to work, to subtly access information on a mixed-reality display while in the grocery store, or to dismiss a phone call while driving. Correspondingly, there is a need to be able to reject all EMG activity when a user is not intentionally engaging with the system (i.e. performing activities of daily living (ADLs)) while enabling users to quickly and intuitively invoke control commands when desired.

While some approaches (mostly limited to offline analyses) are beginning to emerge for addressing the out-of-set input problem for myoelectric control [8–13], this work specifically explores the idea of wake gestures. Not unlike widely familiar wake words in speech recognition, like ‘Hey Siri’ or ‘Hey Alexa,’ the use of EMG-based wake gestures was first introduced by Kumar *et al* [12]. They found that by using a simple dynamic time warping (DTW)-based approach that considered the entire temporal sequence of a discrete gesture, they could reliably reject other ‘outlier’ gestures while maintaining adequate sensitivity to recognize the target gesture. The issue, however, is that this study was limited to highly constrained offline data and the system was not tested in the presence of actual ADLs where significant overlaps between gestures and

activities are more likely [14]. This makes the rejection of ADLs (compared to other unseen gestures) a much more difficult problem as many ADLs involve very similar gestures to what is typically used for control tasks (e.g. opening a door knob involves both closing the hand and pronating/supinating the wrist). Moreover, since there was no user-in-the-loop feedback, the impact of such EMG-based wake gestures on user experience was not explored.

Although wake gestures have had limited exploration in myoelectric control, they have been studied in the context of other input modalities. For example, the work by Pomykalski *et al* explored the social acceptability and user preference of wake gestures for smart assistants and found that the ‘finger snap’ gesture was most preferred among participants [15]. However, this initial work, which leveraged the leap motion controller to detect each gesture in a wizard-of-oz experiment, did not evaluate the ability of this gesture to reject other non-targeted activities. Wrist-flex, a pressure-sensor-based wrist wearable, acknowledged the need for a finger-spread wake gesture; however, it never evaluated its resilience to everyday ADLs [16]. Similarly, Tongue-n-Cheek also acknowledged the need for a wake gesture for tongue gesture recognition [17]. However, their evaluation protocol did not include any intentionally introduced activities that would likely cause false activations for the system, potentially leading to higher perceived performance than would be likely under more realistic scenarios. While WearSign, a sign-language recognition system, evaluated their inertial measurement unit (IMU) raise-to-wake gesture in a pilot study to ensure its robustness, the performance of the wake gesture was computed offline by recording a set of ADLs from each participant, including running, walking and picking up objects [18]. Correspondingly, while many works have acknowledged the importance of wake gestures, their online real-world performance is often an afterthought to the main study.

To evaluate the real-world online use of EMG-based wake gesture-enabled myoelectric interfaces, 20 individuals with varying degrees of experience with EMG engaged in multiple ADLs, including walking, writing, driving, typing, and phone use. While performing the ADLs, participants were randomly cued to complete a secondary dedicated control task (either using a gesture to dismiss an alarm or controlling a remote-controlled robot around an obstacle course). The proposed system ignored >99.9% of all non-target EMG activity during these tasks while reliably recognizing the wake gesture when intentionally activated. These results suggest the use of wake gestures as a promising path forward for enabling the use of myoelectric control as a robust general-purpose ubiquitous input modality that can be intermittently activated.

2. Methods

2.1. Participants and hardware

As approved by the Research and Ethics Board at the University of New Brunswick (on file as REB 2022-122), 20 participants (between the ages of 20 and 31) were recruited for this study. The participants had a range of experience with myoelectric control—with most having little to none. The experiment lasted approximately 1.5 h, and participation was entirely voluntary, with no rewards or incentives provided. This study was conducted in full compliance with the ethical principles outlined in the Helsinki Declaration.

All data in this experiment were recorded using the Myo Armband (see figure 1), a previously commercially available surface-EMG wearable device recording EMG at 200 Hz across eight channels. The Myo Armband was chosen due to its convenience and its previous use within the consumer market. At the start of the session, participants placed the armband on their dominant forearm, at approximately 1/3 of the length, proximal to the elbow. EMG data were recorded for the entire session to enable the post hoc analysis of the system's performance.

2.2. Wake Gesture enabled myoelectric interface

In this work, the myoelectric interface can be defined by two distinct pieces. First, there is a *dedicated myoelectric control system* that, when active, classifies between a closed set of gestures. This system is representative of any general-purpose myoelectric control system such as the handwriting recognition system explored in [7], a simple classifier to scroll through a menu in mixed reality [19], or a drone controller [20]. As previously described, these dedicated systems will be prone to false activations if not turned off when users engage in other ADLs. Correspondingly, to enable users to easily and quickly toggle the system into active and sleep states, there is a second independent system, the *wake gesture recognizer*, which can toggle the state of the dedicated classifier. A further description of each is presented as follows:

2.2.1. Dedicated myoelectric control system

For each participant, a personalized, dedicated continuous myoelectric control model was trained to discriminate between five static gestures: wrist flexion, wrist extension, hand open, hand close, and no movement (Rest). These gestures were chosen as they were also employed by the Myo Armband and are widely adopted in research [21]. This was the underlying classification model used for control within both tasks highlighted in section 2.4. To train this model, participants were prompted to elicit the displayed gesture in a traditional screen-guided training session (i.e. where users follow along with images on a screen indicating which contraction they should be eliciting) [22]. Five repetitions of moderate-intensity ramp



Figure 1. The Myo Armband is a previously commercially available 8-channel, 200 Hz, surface-EMG wearable device.

contractions (i.e. contractions starting from rest) of three seconds each were captured for each of the five input classes.

While the underlying dedicated classifier was a linear discriminant analysis (LDA) model, a simple machine learning model commonly used for prosthesis control [23], it could have been any closed-set model (either discrete or continuous as defined in [1] or even a regression-based system [24]). All EMG data were split into windows of size 200 ms with 100 ms increments, resulting in ten classifier predictions per second. Hudgins' time domain feature set (i.e. mean absolute value, zero crossings, waveform length, and slope sign changes) [25] was used to describe each window of EMG for classification. A proportional velocity control scheme was also adopted, meaning that each class prediction was accompanied by an associated speed based on the intensity of the contraction, as is common for prosthesis control [26].

2.2.2. Wake gesture recognition system

The wake gesture recognition system, which is independent of the aforementioned closed-set model, employed a simple DTW approach similar to that proposed in the offline study by Kumar *et al* [12]. This approach was used as it enables the recognition of discrete gestures (with underlying temporal dynamics) by evaluating the Euclidean distance between two multi-variate time series (i.e. templates) after accounting for different warpings [27]. Additionally, it has been shown to have better wake gesture detection performance than non-temporally encoded algorithms [12] and has even been shown to outperform more complex deep models for within-participant classification tasks [28]. As highlighted below, 20 ground-truth discrete templates were recorded for each participant and used as target wake gesture templates. Each template (one second of raw EMG data) was split into windows of 150 ms with 50 ms increments as proposed in [12] and confirmed via pilot studies. Compared to the dedicated classifier from above, the shorter window length and increment were chosen to improve the temporal resolution of the DTW. Finally, the root mean square feature was extracted from each of the eight channels for each window.

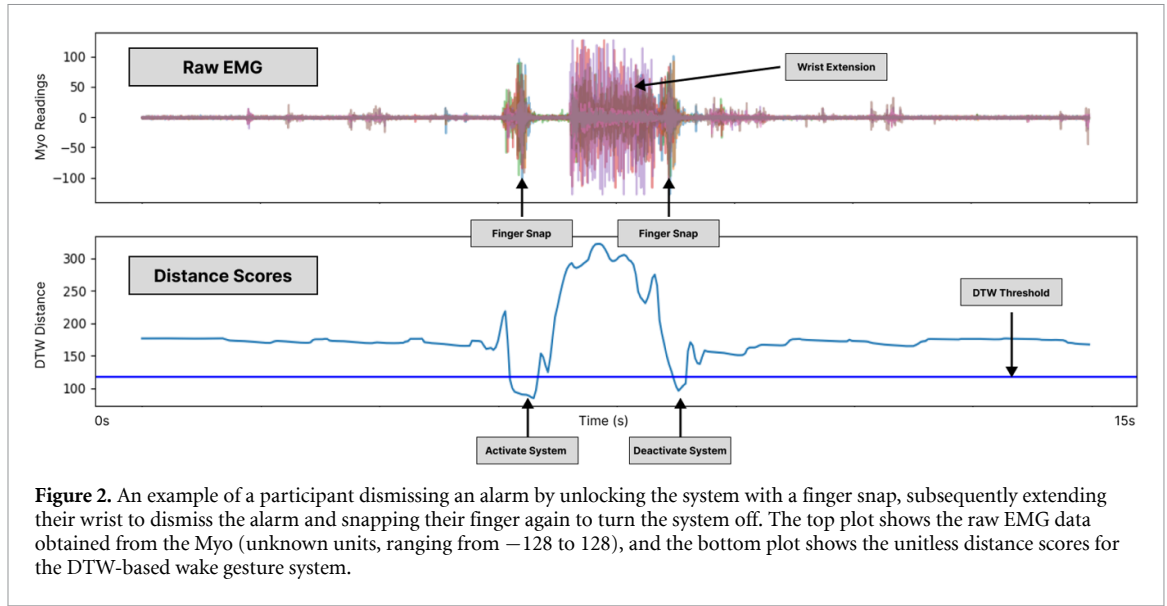


Figure 2. An example of a participant dismissing an alarm by unlocking the system with a finger snap, subsequently extending their wrist to dismiss the alarm and snapping their finger again to turn the system off. The top plot shows the raw EMG data obtained from the Myo (unknown units, ranging from -128 to 128), and the bottom plot shows the unitless distance scores for the DTW-based wake gesture system.

The threshold used for the DTW was computed as follows:

$$D = \sum_{i=1}^N \text{dist}(t_1, t_2) \quad (1)$$

where N is the unique combination of all two templates, t_1 is one of the twenty templates, t_2 is another of the unique templates, i is the current unique pair of templates, dist is the DTW alignment cost between two multivariate time series, and D is the sum of distances of all unique template combinations.

$$T_{\text{thresh}} = \frac{D}{N} + s\sigma \quad (2)$$

where T_{thresh} is the wake gesture model's threshold, σ is the standard deviation of the vector D , and s is the number of standard deviations away from the mean distance between templates allowed.

For the wake gesture recognition system, one-second templates were passed through the DTW model every 50 ms using a sliding window. The average distance of each new template to all of the gesture templates in the training set was computed and compared to the tuned threshold T_{thresh} . If the average distance between the new template and the training set was below the predefined threshold, the template was classified as a finger snap gesture (toggling the control mode of the dedicated classifier between sleep and input, thus turning the system on or off). A majority vote was then taken over five subsequent predictions (250 ms) to further improve the algorithm's robustness to false activations and to better account for template alignment. An example decision stream of the thresholds of incoming EMG templates is shown in figure 2. An activation and deactivation audio stimulus was provided (similar to the generated sound when plugging in and removing a USB stick from

a computer) to indicate when they had successfully activated and deactivated the system.

The finger snap gesture was selected based on previous wake gesture work for computer vision [15], initial pilot studies, and because finger snaps are inherently dynamic, playing into the strengths of the temporal nature of the DTW approach used for detection. Following data acquisition, participants were guided through a one-minute unstructured session where mock ADL data were captured to obtain the ideal distance threshold to tune the wake gesture model (see equation (2)). During this session, participants were asked to elicit gestures that might incorrectly activate the system, such as tapping and waving their fingers. This data was then used to generate the false positive rate (FPR) of a receiver operating characteristic (ROC) curve. The point nearest the top left corner of the ROC curve that still had no false positives was chosen as the threshold, as this represented the model's best true positive rate (TPR) (its ability to recognize the training templates as finger snaps) while ensuring a FPR of zero (its ability to ignore all data recorded during the one minute of mock data).

2.3. ADLs

After training the two separate models to (1) recognize the wake gesture and (2) classify the five continuous control gestures, the combined system was evaluated during a representative set of ADLs. The activities chosen to evaluate the control system came from conversations with pilot participants about the activities they spend the most time doing throughout the day. While these tasks do not represent all possible variations of everyday activities, they were meant to encompass various tasks when EMG-based inputs may be useful (e.g. a music player while walking or a gesture-based keyboard augmentation). Two minutes of data were recorded for each activity, and each activity was repeated during both control tasks

(see section 2.4) for a total of 20 min ($2 \text{ min} \times 2 \text{ tasks} \times 5 \text{ activities}$) of ADL data per participant. These activities were presented randomly to the participants while testing each of the control tasks. Participants' actions were not constrained during the experiment to ensure the most real-world data possible. For example, although participants were told to engage in activities such as walking or driving, their other natural movements, like scratching their nose, cracking their fingers, putting their hands in their pockets, and pushing on the chair's armrest when standing up, were not constrained in any way. Although these behaviours were not necessarily expected when the study protocol was designed, they led to a more realistic collection of ADL. The following is an overview of the five ADLs evaluated in this study:

- **Walking:** participants were asked to walk on a stationary treadmill at a slow walking speed of 2 km hr^{-1} . While this ADL did not necessarily promote active muscle movements, it was included to ensure that individuals could interact with the system (including toggling its state) as they continued to engage with the activity.
- **Writing:** participants were given a book, paper, and a pen and were asked to write a passage while occasionally changing pages.
- **Typing:** participants were asked to type at a comfortable speed and perform an untimed typing test³.
- **Driving:** participants used a Thrustmaster TMX⁴ force feedback driving simulator (which vibrated during driving) and the RaceRoom Racing Experience game to simulate a realistic driving experience.
- **Phone:** participants were asked to complete common phone activities, including responding to text messages, scrolling through social media, and any other phone-related tasks.
- **Other:** participants conducted various other actions, such as standing up from their chairs, scratching their noses, grabbing their phones out of their backpacks, tying their shoes, talking with their hands, etc. These activities were not directly prompted and thus represent additional self-initiated tasks that further tested the resilience of the wake gesture algorithm.

2.4. Myoelectric control tasks

The proposed wake gesture-governed myoelectric interface was evaluated in two unique and fundamentally different tasks. These tasks were evaluated in the context of each ADL presented above. The first task involved a complete context switch as participants navigated a robot around an obstacle course. The second task represented the ubiquitous

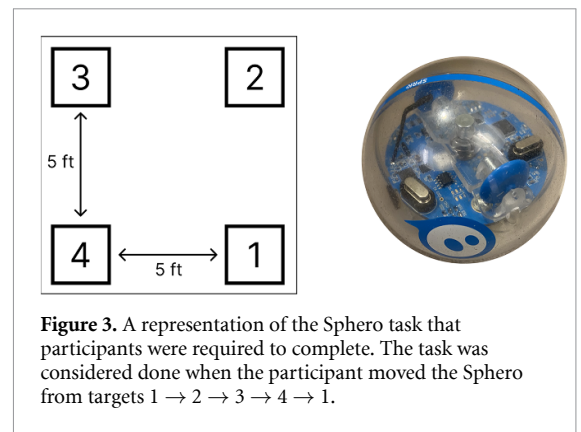


Figure 3. A representation of the Sphero task that participants were required to complete. The task was considered done when the participant moved the Sphero from targets $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$.

use of myoelectric control and involved dismissing an alarm. Users were periodically prompted to address each task, with notifications recurring four times within each two-minute ADL at 30 s intervals. When prompted, users would stop the activity to engage with the particular task (e.g. they would stop typing and begin controlling the robot). It is important to note that we were not concerned about participants' performance within the dedicated control tasks nor with the control enabled by the dedicated classifier, as this study focused on the wake gesture recognition system. Correspondingly, the tasks were meant to serve as 'mock' environments to evaluate the intentional activation of the wake gesture.

2.4.1. Task 1: Sphero

The first task was the control of a Sphero—a spherical robot that can roll across the ground at various speeds. In this task, participants had to control the robot around a 5×5 foot course (shown in figure 3). After travelling from spot $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$, the task was considered complete. Note that this course had to be modified for two participants. One suffered from muscle fatigue due to an underlying injury unrelated to the experiment and one had challenges eliciting a repeatable hand-open contraction. The mapping from gestures to the robot control were hand open (Move away), hand close (Move toward), wrist flexion (Move left), wrist extension (Move right), and no movement (Do nothing). The flexion and extension commands were reversed for left-handed users to improve intuitiveness. When prompted, the participants stopped engaging in the ADL being tested, walked over to the course, snapped to turn the robot on, navigated it through the obstacle course, snapped to turn the robot off, and then returned to their respective activities. In this case, 'stopping' refers to the complete context switch from the ADL (i.e. they would stand up out of their chair and walk over to the Sphero course). The only activity the user continued to perform during control was walking on the treadmill. This meant that participants continued walking on the treadmill as they controlled the Sphero. During

³ <https://official-typing-test.com/test/untimed.html>.

⁴ www.thrustmaster.com/en-us/products/tmx-force-feedback/.

this task, if the wake gesture system was unintentionally activated, the robot would turn on and would have to be deactivated before the user could continue engaging in the particular ADL.

2.4.2. Task 2: Alarm deactivation

The second task was simpler and involved dismissing an alarm. Participants snapped to activate the system, extended their wrist to silence the alarm, and then snapped to deactivate the system. Although easier and only requiring a few seconds, this task aimed to simulate a more ubiquitous use case of system control where participants were not required to completely switch contexts. This meant users dismissed the alarm without switching contexts from the activity in which they were previously engaged. For example, when driving, they would take one of their hands off of the wheel or when writing, they would either put the pencil on the desk or switch it to the other hand momentarily.

2.5. Data collection, interview, and survey

A set of seven other potential wake gestures was collected at the end of the data collection to evaluate the offline performance of the DTW-based approach when presented with real-world ADLs (something that was not explored in the work by Kumar *et al* [12]). These gestures were hand open + wrist pronation (pressing a 'button'), finger gun (with recoil), hand open (pulse), double middle finger tap, wrist roll (pronation then supination), wrist extension (flick right), and wrist flexion (flick left). Ten repetitions of each gesture were acquired. Finally, participants completed a Likert-style survey, where their experience with the system was assessed alongside supplementary informal interviews. In these unstructured interviews, participants were asked how the system could be improved and about their preferences on the tradeoff between true positives and false negatives [29].

3. Results

3.1. Non wake gesture approach

The distribution of unintended class activations that would have occurred during the ADLs if no wake gesture was used to toggle the dedicated classifier is shown in figure 4–Left. This was computed post hoc by running the dedicated classifier (outlined in section 2.2.1) on the ADL data recorded when not actively controlling the Sphero during Task 1. These results suggest that when performing the ADLs, the dedicated classifier incorrectly predicted that the participants were eliciting one of the control gestures (i.e. wrist flexion, wrist extension, hand open, or hand close) the majority of the time. This further exemplifies that such a system (relying only on the no movement class for a do-nothing command, as is typical in prosthetics) would have inevitably resulted in

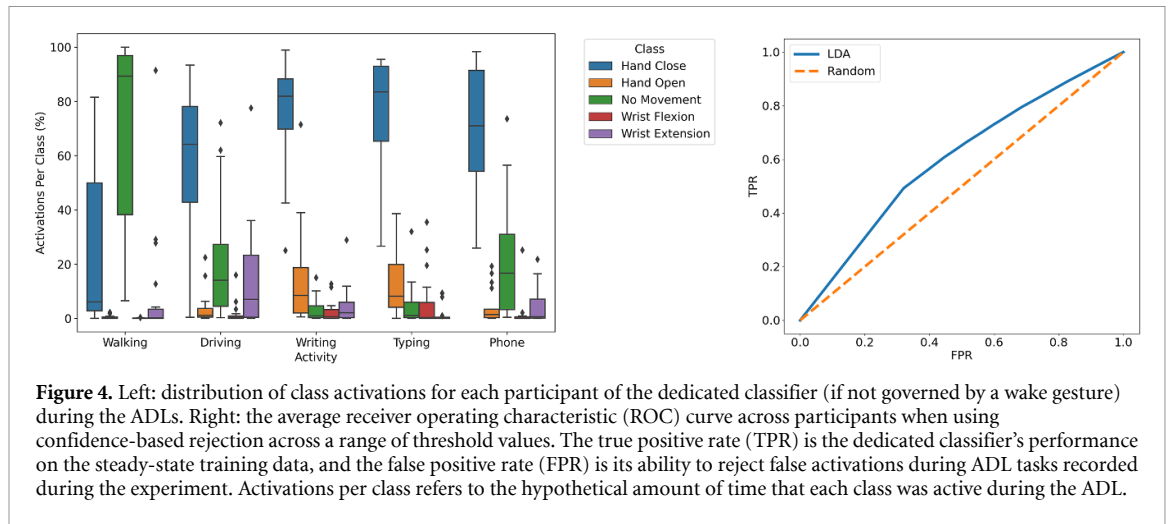
spurious unintended activations and an unusable system. This was consistent across most ADLs, with the exception of walking (which required no use of the hands). In theory, this would have meant that the Sphero robot would have been sporadically moving all around the room.

While some EMG researchers have acknowledged that myoelectric control is an open-set problem through concepts like rejection (i.e. rejecting classifier outputs that fall below a predefined confidence threshold) [30], our results show that such techniques do not work for ignoring out-of-set inputs during ADLs performed by able-bodied users, as highlighted in figure 4–Right. Here, the dedicated continuous classifier's ROC curve is shown, emulating the results that would have been seen if a confidence-based rejection scheme had been used. This plot suggests that a naive rejection approach is only slightly better than randomly guessing if a given input should be rejected. This supports previous research that highlights the issue of overconfidence with closed-set myoelectric control models. [11, 31] These results reinforce the need for specialized strategies to deal with inadvertent muscle-based inputs when leveraging myoelectric control for applications outside of prosthetics.

3.2. Wake gesture system performance

The number of true positives, false positives, and false negatives were recorded throughout the experiment to evaluate the online performance of the wake gesture system. False positives were counted whenever the system state was toggled unintentionally (i.e. the snap gesture was falsely recognized). In contrast, false negatives were manually counted whenever an individual elicited a snap, and the system did not respond. Due to the sound of the snap gesture, this was a relatively trivial process. The number of true negatives (i.e. correctly rejected activity) was determined post hoc by evaluating the total number of decisions made by the wake gesture system on the recorded data from each task.

The confusion matrices presented in figure 5 denote the *false negatives*, *false positives*, *true positives*, and *true negatives*. The number of true positives was consistent by design across participants and tasks ($4 \text{ per activity} \times 5 \text{ activities} \times 2 \text{ toggles (one for on and one for off)} = 40$). The near-zero FPR indicates the goal of effectively eliminating inadvertent activations while maintaining reasonable TPRs of 75.1% and 81.1% for the two respective tasks was achieved. A non-parametric Wilcoxon signed-rank test was run to check for statistical significance between the number of false negatives between the two tasks. The test results revealed a significant difference ($z = 2.56$, $p = 0.011$), indicating that participants had more difficulty replicating the wake gesture template during the alarm task. It is hypothesized that this was because there was less of a context switch compared to the Sphero task.



True Label	Task 1: Sphero		Task 2: Alarm	
	Other	Snap	Other	Snap
	Predicted Label	Predicted Label	Predicted Label	Predicted Label
Other	(100.0%) 30923 ± 3796	(0.0%) 0.1 ± 0.1	(100.0%) 13986 ± 1548	(0.0%) 0.0 ± 0.0
Snap	(18.9%) 9.3 ± 9.3	(81.1%) 40.0 ± 0.0	(24.9%) 13.3 ± 10.5	(75.1%) 40.0 ± 0.0

Figure 5. A confusion matrix for each task, showing the predicted and true labels for each decision during the online evaluation of the wake gesture system. The top value in each box is the average percentage across participants, and the bottom value is the actual number of events (with the standard deviation) averaged across participants.

The results of two subjects were withheld from the confusion matrices and their respective analyses as they were deemed outliers. The first outlier (z -score > 3) had 141 false negatives, indicating this participant far exceeded the 99.6 percentile of the distribution observed across the other subjects. Further, there was justification for their removal as they reported having excessive muscle fatigue toward the end of the experiment. This outcome was thought to have been induced by the consistent activation and deactivation of the system, which would be atypical for a realistic intermittently used myoelectric interface. The second outlier (z -score > 3) had seven false positives, which was likely due to the threshold for this participant being inadvertently set too low during the initialization of the wake gesture algorithm, failing to properly favour false negatives over false positives. It is hypothesized that this would not have occurred if an adaptive approach that enabled continuous tuning throughout the experiment had been implemented.

3.3. Evaluating other wake gestures

To evaluate the offline performance of this approach to other wake gestures, the post hoc performance

of seven other potential gestures acquired at the end of the study was evaluated by training an off-line wake gesture recognition system (as presented in section 2.2.2) and running it over all of the ADL data acquired during both tasks. Figure 6 shows the area under the ROC curve (AUC) percentage across participants. The AUC is the integral of the ROC curve for a single participant, resulting in a value between zero and one, indicating the model's ability to accept deliberate activations and reject inadvertent input. A perfect wake gesture system would have an AUC percentage of 100%, indicating that it can ignore all ADL data while accepting all the training gesture templates. A non-parametric Friedman test with a Finner posthoc comparison was used to check for significant differences between the AUCs of the different gestures, with the only significant difference being found between the button press and wrist extension gestures ($p < 0.005$). Unfortunately, it was difficult to ascertain the offline performance of the finger-snap wake gesture due to intentional activations during the ADL tasks. Therefore, while we cannot directly claim that this DTW-based approach generalizes to other wake gestures, offline performance suggests they might also be robust to ADL. Future work should confirm this by letting the users define their own unique wake gestures.

3.4. User survey and interview

To capture participants' feelings on using EMG-based wake gestures, they completed a Likert-style questionnaire, presented in figure 7. In general, the perception of the system was positive despite the intentionally high levels of rejection to reduce the potential of false activations. The most critical participant (the 'strongly disagree' for the accidental triggering of the wake gesture) was the outlier participant mentioned above, who still only had seven false positives in the entire experiment, further highlighting the detriment of false activations on user experience. Through informal interviews with participants, there

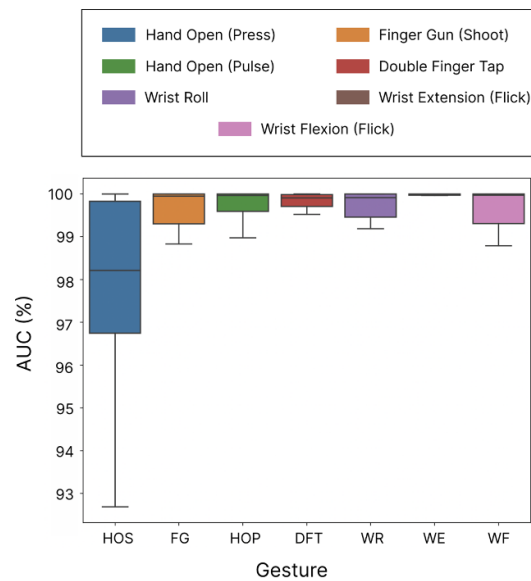


Figure 6. The area under the curve (AUC) percentage when using each wake gesture (recorded at the end of the online experiment) and running the system post hoc on the ADL data. Note that the outliers have been removed for visualization purposes.

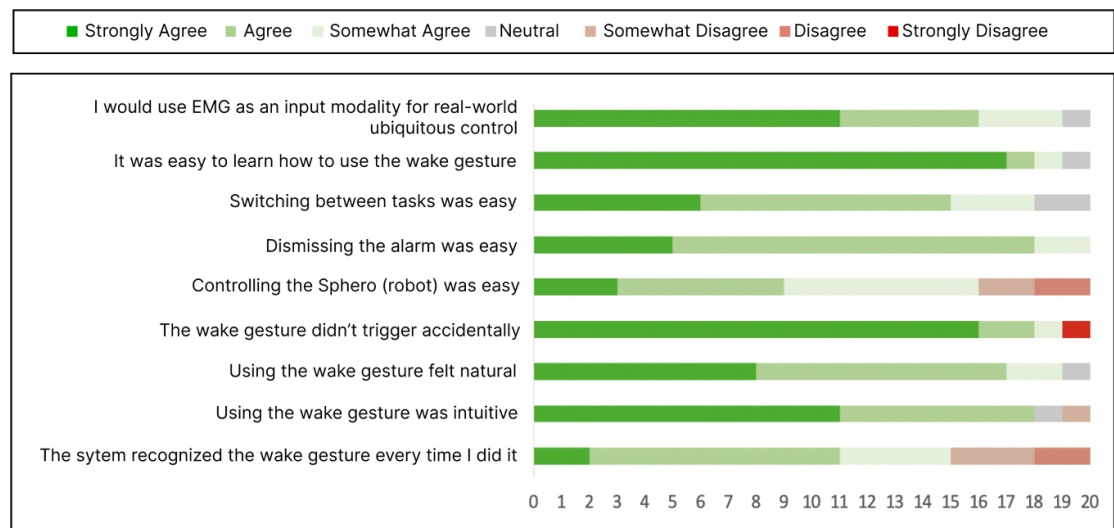


Figure 7. The results from the seven-point Likert-style questionnaire evaluating user perception of the wake gesture governed myoelectric control interface.

was a consensus that a wake gesture system should employ strict rejection settings to ensure no false positives, as dealing with erroneous inputs requires more effort from the user than just quickly re-eliciting the wake gesture. Moreover, it was generally agreed that false negatives were not overly cumbersome, with one participant even noting that having to redo the wake gesture once was not a big deal since snapping is ‘so quick and easy’. Another participant described tuning toward eliminating false positives as ‘the casing on a big red button’ (i.e. it was an added protection against undesirable consequences), emphasizing that

they preferred the extra protection against the possibility of accidental activations. However, it should be noted that many participants commented that the more false negatives that occurred in succession, the more frustrated they became. Another participant questioned the social acceptability of the snap gesture due to the noise it makes, highlighting the potential use of other, more subtle wake gestures. In general, however, the consensus was that the snap wake gesture was intuitive and natural, with one participant even suggesting that it was an easier way to activate a system than voice-activated smart home systems,

evoking interesting possibilities for potential multi-modal interaction systems.

4. Discussion

This work proposes, evaluates, and sets a baseline for using wake gestures to eliminate false activations for myoelectric interfaces. The resilience of the proposed wake gesture approach to inadvertent activations was demonstrated during everyday activities, including writing, walking, typing, driving, and phone interaction. A traditional dedicated closed-set classification approach would not have worked during these tasks due to the constant unintentional recognition of muscular inputs. The proposed template-based approach for recognizing a discrete finger snap to toggle the dedicated control system effectively reduced false activations. In general, it could ignore *all inadvertent activations* during the set of ADLs while maintaining adequate recognition rates of the wake gesture to enable reliable, always-available ubiquitous input. While this work provides a solid foundation for the future of intermittent general-purpose myoelectric control, more work is needed to advance this technology to a widely adopted ubiquitous input modality. With this in mind, this section critiques this work and highlights future research directions to better characterize its contributions.

4.1. The tension between false negatives and false positives

For any open-set classification problem, the ability of a system to ignore inadvertent activations (i.e. out-of-set inputs) will always be at odds with its ability to recognize true positives (i.e. within-set inputs). Improving (or decreasing) the false negative rate will increase the number of false positives and vice versa unless the underlying separability and repeatability are improved (e.g. better features, more sensors). While the wake gesture system proposed in this study achieved a median FPR of 0.0%, this came at the cost of an increased false negative rate (18.9% and 24.9% in the Sphero and alarm tasks, respectively). This meant that approximately one out of five times, users had to evoke the wake gesture more than once to successfully toggle the system. Although seemingly undesirable, post-study interviews with participants confirmed that these false negatives were much preferred to inadvertent activations, corroborating previous work evaluating the tradeoff between the two [29]. Interestingly, this was noted even though 18/20 participants were not impacted by false positives during the experiment, as the control schemes were deliberately tuned this way.

A significant hindrance discussed by many of the participants, however, was when they were required to repeat the wake gesture more than a few times before it was successfully recognized. This highlights the need for user-adjustable thresholds or adaptive

algorithms that actively detect successive intentional activations (and potentially temporarily increase the threshold), reducing the number of consecutive false negatives. Additionally, future work could improve the system by adaptively incorporating new templates during regular use [32–34]. These could include templates obtained from successful activation attempts or even from unsuccessful attempts that are closely followed by a second attempt. Generally, however, now that a baseline has been set, future work should focus on increasing the TPR while maintaining a FPR of 0%.

4.2. Other applications for wake gestures

One seemingly promising use of wake gestures could be during prosthesis control, where mechanical locking mechanisms are still widely used to achieve similar effects [35, 36]. Such a wake gesture approach would enable amputees to selectively toggle their prostheses into an active state when desired and an inactive state when performing activities where their prostheses are not needed. For example, a relatively simple and easy-to-remember double or triple pulse of the flexor could unlock and wake up the prosthesis. This means that an individual would not have to worry about pressing a mechanical button, which might not be possible if their non-prosthesis hand is full or already engaged. Alternatively, amputees could lock their prostheses into specific grasps without worrying about spurious activations [37, 38]. For example, one could position their prosthesis into a grasp around a warm cup of coffee and then subsequently elicit the predefined wake gesture to lock the grasp, ensuring that the hand will not inadvertently open. Future work should evaluate the efficacy of wake gestures in these settings.

Even beyond myoelectric control, this work is generalizable to any ubiquitous input modality in which false activations could occur. Novel input modalities, including myoelectric control, are often evaluated in highly constrained settings. However, any interaction techniques that rely on sensed inputs that do not have a general default inactive state (i.e. that have to worry about inadvertent activations) will struggle to find adoption within real-world use cases. For example, although researchers can achieve high recognition accuracy on IMU-based gesture recognition tasks like air-drawing [39] and toe-based inputs [40], how can these systems be enabled in real-world scenarios when engaging in other tasks? As highlighted in this work, optimizing offline recognition and evaluating these modalities in constrained settings is inadequate if the goal is to progress these technologies toward widespread consumer adoption. Correspondingly, emerging systems need to be stress-tested within realistic settings [41], or the gap between offline performance and online usability may continue to grow [42, 43].

4.3. Algorithmic improvements and advancements

This work evaluated a simple time series distance measurement (DTW) to measure the distance between target templates and potential wake gestures. Future work should explore other algorithms (e.g. deep learning) or more robust features [44] to improve the resilience of wake gesture systems to such factors [7, 11]. One interesting thing to note, however, is the importance that DTW played in considering the temporal profile of the gesture rather than traditional machine learning approaches that assume stationarity. For example, when the temporal structure of wrist flicks was leveraged (extension and flexion), they could be actively discerned from the ADL data (i.e. the AUC was high). This was not the case when using their static (non-dynamic) counterparts, where the LDA classifier could not differentiate these inputs from the ADLs (at least in the context of the closed-set dedicated control classifier). This suggests that the inherent dynamics in discrete gestures are crucial for separating target gestures and non-target activities, particularly in the context of myoelectric control. These results highlight a promising path forward for using more temporally defined discrete gestures within HCI applications [1, 7].

Another outcome was the significant difference in TPR between the Sphero and Alarm tasks. It is conceivable that without switching contexts, users were more distracted when eliciting the wake gesture, thus leading to less repeatable gestures. Future work should consider how to train algorithms to become resilient to such confounding factors through real-time adaptation or more representative data collections. Finally, it should be noted that the Myo Armband, which was used in this study, did previously use an unlock gesture (a double-finger tap) and an automatic two-second timeout to enable moded entry into the underlying gesture recognizer. Many users, however, found that the Myo approach lacked the requisite robustness for real-world use [45–47], with many indicating that it was overly sensitive, lacked reliability, and was sometimes challenging to trigger [48–50]. Additionally, this device is now discontinued, and its design, implementation, and evaluation were never externally validated or publicly released, meaning there is currently no way to evaluate or baseline the wake gesture approach Myo took. Moreover, the initial wake gesture work done by Kumar *et al* was limited to offline data and was never tested in the presence of ADLs. Correspondingly, this is the first online real-world study that evaluates the use of EMG wake gestures to ignore unintentional muscle activations.

4.4. Limitations

Although this study evaluated a range of common ADLs, it is uncertain how this approach might generalize to other, potentially more active tasks. For example, the act of walking is unlikely to generate

EMG that would be confused, but activities that are common during walking (such as scratching one's nose or wiping one's forehead), may. Future work should evaluate these wake gesture approaches in the presence of these types of activities, including more complex object manipulations, such as while cooking or eating. Similarly, this approach should be evaluated in more dynamic activities (e.g. at the gym while doing push-ups) and in rapidly changing environments with a lot of context switching. Additionally, this work specifically explores within-session user-dependent models. For the eventual success of this type of approach, it will be important that they work across sessions and, potentially, across users. While recent works show that this will likely be possible when there is enough data [7, 11], there is still a need for large datasets consisting of ADLs. Finally, it could be interesting to explore using wake gestures in the context of hands-full activities. For example, how could a tap on a pencil or a clench on the steering wheel enable interaction? Such approaches that reduce the need for a free hand could greatly improve myoelectric interfaces and the applicability of wake gestures.

5. Conclusion

This work takes a meaningful step toward the real-world viability of myoelectric control as a widely applicable ubiquitous input. In particular, the results suggest that wake gestures are a viable solution for enabling intermittent myoelectric control, effectively eliminating inadvertent activations during everyday ubiquitous tasks. In an online evaluation, near-perfect performance in rejecting unintended device inputs while maintaining adequate robustness for two myoelectric control tasks was achieved (dismissing an alarm and controlling a robot). Moving forward, researchers should consider using wake gestures when designing interactive systems, particularly when using sensed inputs that may be inadvertently activated during unrelated ADL.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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